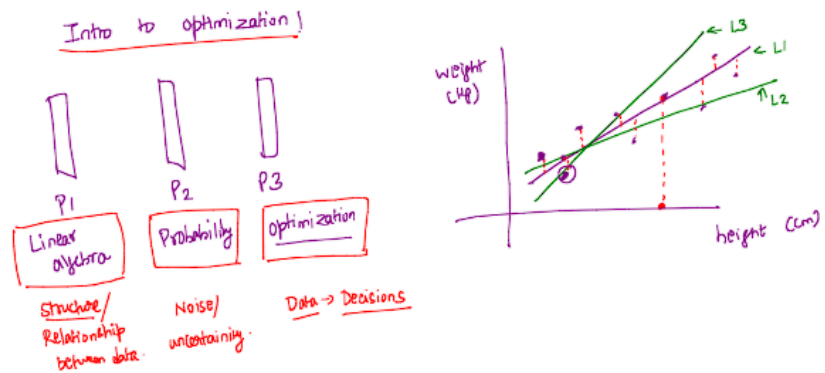


MLF Week 8 : Unconstrained Optimisation

Lecture 1 : Pillars of machine learning

TDR

5/22/2021



Lecture 2 : Introduction to Optimization

Why optimisation?

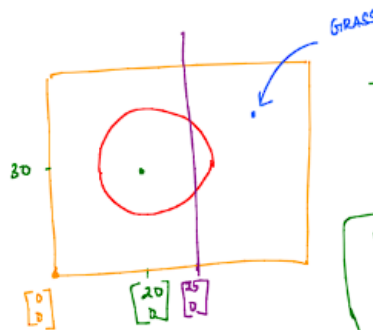
→ We care about finding the "best" classifier!

"Least" Loss

"maximum" reward

Example

- A cow is at $\begin{bmatrix} 20 \\ 20 \end{bmatrix}$
- Cow tied to a 10 unit radius rope.
- Perpendicular fence that passes through $\begin{bmatrix} 25 \\ 0 \end{bmatrix}$



- Grass on the field at $\begin{bmatrix} 40 \\ 40 \end{bmatrix}$

"How close can the cow get to the grass?"

What are we measuring?

- Distance from grass $\begin{bmatrix} 40 \\ 40 \end{bmatrix}$

Say cow is at $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$d\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \begin{bmatrix} 40 \\ 40 \end{bmatrix}\right) = (x_1 - 40)^2 + (x_2 - 40)^2$$

What to do with distance?

- As "small" as possible.

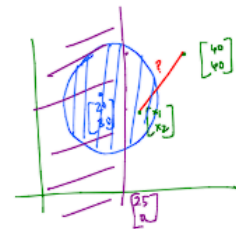
$$\min_{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} (x_1 - 40)^2 + (x_2 - 40)^2$$

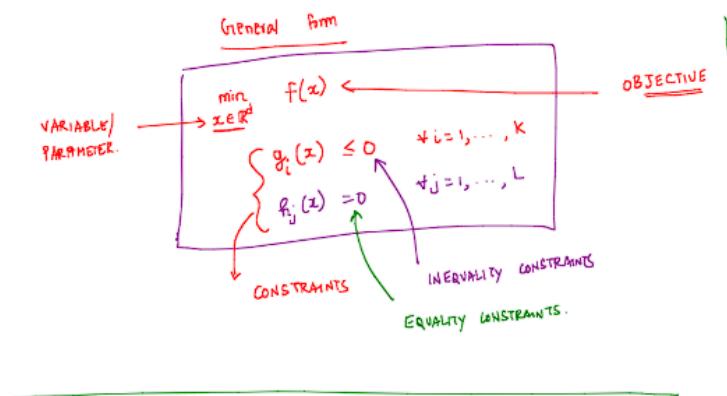
Rope restriction

$$(x_1 - 20)^2 + (x_2 - 20)^2 \leq 10^2$$

Fence restriction

$$x_1 \leq 25$$





UNCONSTRAINED OPTIMIZATION

$$\min_{x \in \mathbb{R}} (x-5)^2 = f(x)$$

Solution $x=5$ Objective value at $x=5$ $= 0$.

$$f'(x) = 2(x-5) = 0$$

$$\Rightarrow \boxed{x^* = 5}$$

$$\min_{x \in \mathbb{R}} 3x^6 + 2x^5 + 3x^3 + 5x^2 + 2$$

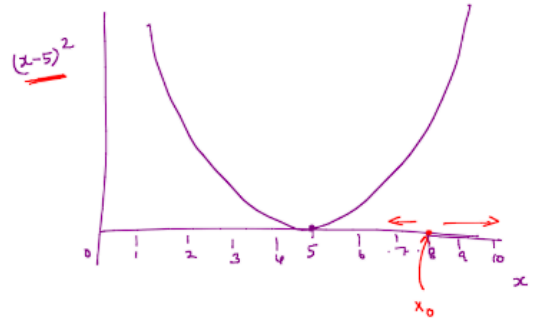
$$f'(x) = 18x^5 + 10x^4 + 9x^2 + 10x = 0$$

Lecture 3 : Solving an unconstrained optimization problem (Part 1)

$$\min_{x \in \mathbb{R}} (x-5)^2$$

Approach:-

- Start with $x_0 \in \mathbb{R}$ [arbitrary choice]
- for $t=1, \dots, T$
- update $x_{t+1} = x_t + d$ direction
- end.



"good direction to move"

$$f(x) = (x-5)^2$$

$$f'(x) = 2(x-5)$$

$$x > 5 \Rightarrow -f'(x) < 0$$

$$x < 5 \Rightarrow -f'(x) > 0$$

CHOOSE $d = -f'(x)$

Given x , what is a good direction?

if	$x > 5$	then	$\leftarrow \begin{matrix} \text{want} \\ d < 0 \end{matrix}$
	$x < 5$	then	$\rightarrow d > 0$

$\Rightarrow$ direction d will depend on x .
i.e. d must be a function of x

$$f(x) = (x-5)^2$$

$$f'(x) = 2(x-5)$$

$\rightarrow$ $x_{t+1} = x_t + d$ where $d = -f'(x)$

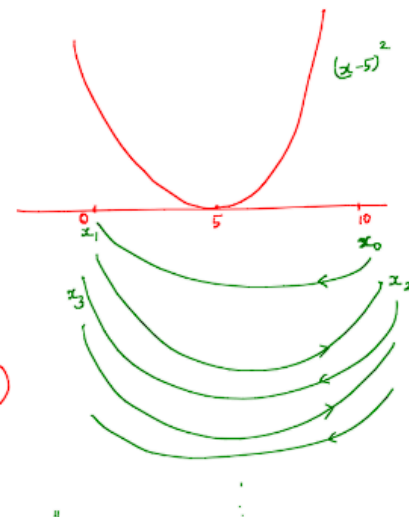
Example

$$x_0 = 10$$

$$d = -f'(x_0) = -2(x_0 - 5) = -2(10 - 5) = -2 \cdot 5 = -10$$

$$x_1 = x_0 + d = x_0 - f'(x_0) = 10 + (-10) = 0$$

$$x_2 = x_1 + d = x_1 - f'(x_1) = 0 + (-2(0 - 5)) = 0 + 10 = 10$$

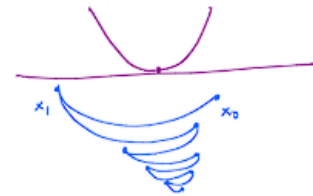


Problem: Not the direction but "amount"

Lecture 4 : Solving an unconstrained optimization problem (Part 2)

$$x_{t+1} = x_t - \underbrace{\eta_t f'(x_t)}_{\text{direction}}$$

STEP SIZE (Scalar quantity positive)



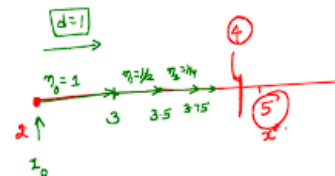
How to choose step size (as a function of t) ?

✗ First attempt:

$$\eta_0 = 1, \eta_1 = \frac{1}{2}, \eta_2 = \frac{1}{4}, \eta_3 = \frac{1}{8}, \dots$$

$$\eta_t = \frac{1}{2^t}$$

No idea how close x_0 is to x^* ← optimal value.



$$\eta_0, \eta_1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$$

$$\eta_t = \frac{1}{t+1}$$

good step size sequence. ✓

$$\sum_{t=0}^{\infty} \eta_t = \sum_{t=0}^{\infty} \frac{1}{t+1} = 1 + \frac{1}{2} + \frac{1}{3} + \dots = \infty$$

$$\eta_t = \frac{1}{2^t}$$

1, 1/2, 1/4, ...

$$\sum_{t=0}^{\infty} \eta_t = \sum_{t=0}^{\infty} \frac{1}{2^t} = 2$$

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 2$$

$$\min_{x \in \mathbb{R}} f(x)$$

$$\min_{x \in \mathbb{R}} f(x)$$

ALGORITHM - GRADIENT DESCENT ALGORITHM.

Initialize at $x_0 \in \mathbb{R}$

for $t=1, 2, \dots$

$$x_{t+1} = x_t - \eta_t f'(x_t)$$

where

$$\eta_t = \frac{1}{t+1}$$

End.

Lecture 5 : Basic algorithm for unconstrained optimization: Gradient Descent

$$\min_{x \in \mathbb{R}} f(x) \leftarrow$$

$$\min_{x \in \mathbb{R}^d} f(x)$$

ALGORITHM - GRADIENT DESCENT ALGORITHM.

Initialize at $x_0 \in \mathbb{R}$

for $t=1, 2, \dots$

$$x_{t+1} = x_t - \eta_t \underline{f'(x_t)}$$

where

$$\eta_t = \frac{1}{t+1}$$

End.

Gradient - descent

Properties

- If $\eta_t = \frac{1}{t+1}$, the algorithm converges
- Gradient descent converges to "LOCAL MINIMUM"

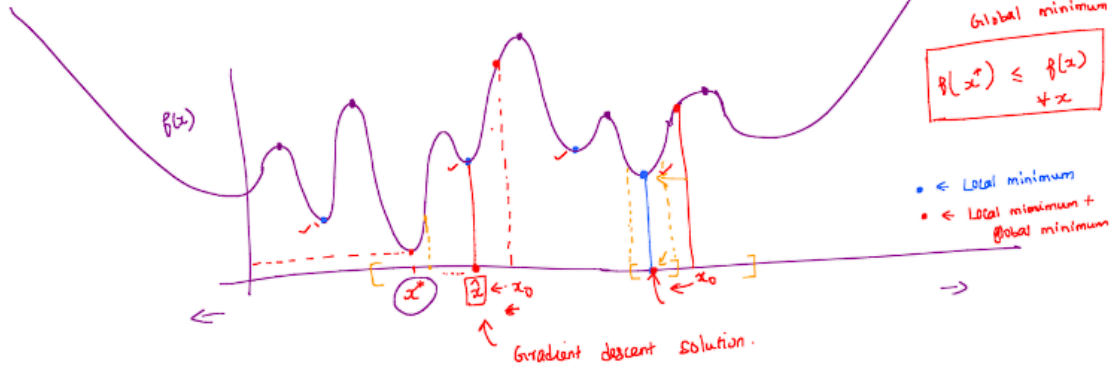
Local minimum

$$\forall \epsilon > 0 \text{ } \exists \delta$$

$$f(\hat{x}) \leq f(x) + x \in [\hat{x} - \epsilon, \hat{x} + \epsilon]$$

Global minimum

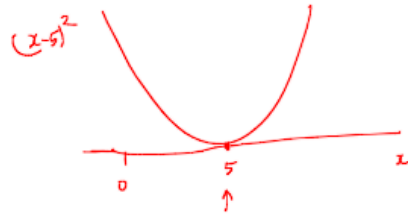
$$f(x^*) \leq f(x) \quad \forall x$$



Functions for which

local minimum
↕
Global minimum

← CONVEX
FUNCTIONS



What is special about

$$d = \boxed{-f'(x)}$$

QN : 1,4

Lecture 6 : Gradient Descent and Taylor Series

8/5/2021

$\min_x f(x)$

Gradient descent

update rule: $x_{t+1} = x_t + \eta \underbrace{(-f'(x_t))}_{\substack{\text{What is so special about} \\ -f'(x)?}}$

Taylor's series: $f(\underbrace{x + \eta d}_{\bar{x}}) = \underbrace{f(x)}_{\text{evaluations are all at } x} + \underbrace{\eta d f'(x)}_{\text{evaluations are all at } x} + \underbrace{\frac{\eta^2 d^2}{2} f''(x)}_{\text{evaluations are all at } x} + \dots$

Local information gives global information

$f(x + \eta d) = f(x) + \eta d f'(x) + \underbrace{f''(x) \frac{\eta^2 d^2}{2} + \dots}_{\text{higher order terms. } \times}$

Small positive step-size η direction

For small enough η

$$f(x + \eta d) \approx f(x) + \eta d f'(x)$$

$\Rightarrow \boxed{f(x + \eta d) - f(x)} \approx \eta d f'(x)$

Function evaluation at updated point along "direction" d

Function evaluation at the current point

Want to choose a direction d s.t

$$\boxed{f(x + \eta d) - f(x) < 0}$$

$\Rightarrow$ Want ' d ' s.t $\eta d f'(x) < 0$

want $\eta d f'(x) < 0 \Rightarrow$ want d s.t. $\boxed{d f'(x) < 0}$

↑
Small
+ve
constant

For the choice of $\boxed{d = -f'(x)}$

$$d f'(x) = -(f'(x))^2 < 0$$

Higher dimensions

$$f(x_1, x_2) = x_1^2 + 4x_2 + 8x_2^2$$

Derivative $\Leftrightarrow$ Gradient
vector of
partial derivatives

$$\nabla f\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \Big|_{x_1=a} \\ \frac{\partial f}{\partial x_2} \Big|_{x_2=b} \end{bmatrix}$$

QN : 3

Lecture 7 : Gradient Descent for multivariate functions

Want $\eta d f'(x) < 0 \Rightarrow$ Want d s.t. $\boxed{d f'(x) < 0}$

$\uparrow$
Small +ve constant

For the choice of $\boxed{d = -f'(x)}$

$$d f'(x) = -(f'(x))^2 < 0$$

Higher dimensions

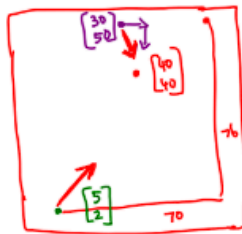
$$f(x_1, x_2) = x_1^2 + 4x_2 + 8x_2^2$$

Derivative $\Leftrightarrow$ Gradient
Vector of partial derivatives.

$$\nabla f \left(\begin{bmatrix} a \\ b \end{bmatrix} \right) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \Big|_{x_1=a} \\ \frac{\partial f}{\partial x_2} \Big|_{x_2=b} \end{bmatrix}$$

$$f(x_1, x_2) = x_1^2 + 4x_2 + 8x_2^2$$

$$\nabla f \left(\begin{bmatrix} 1 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 \\ 4 + 16x_2 \end{bmatrix}_{x_1=1, x_2=3} = \begin{bmatrix} 2 \\ 4 + 16 \times 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 52 \end{bmatrix}$$



$$d(x_1, x_2) = (x_1 - 40)^2 + (x_2 - 40)^2$$

$$\nabla d(x_1, x_2) = \begin{bmatrix} 2(x_1 - 40) \\ 2(x_2 - 40) \end{bmatrix} ; \nabla d \left(\begin{bmatrix} 5 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 2(5 - 40) \\ 2(2 - 40) \end{bmatrix} = \begin{bmatrix} -70 \\ -76 \end{bmatrix}$$

$$\therefore \nabla d \left(\begin{bmatrix} 30 \\ 50 \end{bmatrix} \right) = - \begin{bmatrix} 2(-10) \\ 2(10) \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix} = \begin{bmatrix} 20 \\ -20 \end{bmatrix}$$

$$- \nabla d \left(\begin{bmatrix} 5 \\ 2 \end{bmatrix} \right) = \begin{bmatrix} 70 \\ 76 \end{bmatrix}$$

Gradient
descent

$$\vec{x}_{t+1} = \vec{x}_t + \eta (-\nabla f(x_t))$$

$\uparrow$ vector $\uparrow$ vector $\uparrow$ scalar $\uparrow$ vector

$$f(x + \eta d) = f(x) + \eta d f'(x) + \frac{\eta^2}{2} f''(x) + \dots$$

$\underbrace{\hspace{1.5cm}}_{\text{scalar}}$

Higher order Taylor series

$$f(x + \eta d) = f(x) + \eta d^T \nabla f(x) + \dots$$

$\underbrace{\hspace{1.5cm}}_{\text{vector}} \quad \underbrace{\hspace{1.5cm}}_{\text{vector}} \quad \underbrace{\hspace{1.5cm}}_{\text{scalar}}$

Lecture 8 : Taylor series in Higher dimensions

$$f(x+\eta d) = f(x) + \underbrace{\eta d^T}_{\text{scalar}} \underbrace{\nabla f(x)}_{\text{vector}} + \frac{\eta^2}{2} d^T \nabla^2 f(x) d + \dots$$

Higher order Taylor series

$$f(x+\eta d) = \underbrace{f(x)}_{\text{vector}} + \underbrace{\eta}_{\text{scalar}} \underbrace{d^T}_{\text{vector}} \underbrace{\nabla f(x)}_{\text{scalar}} + \dots$$

$$f(x+\eta d) - f(x) \approx \eta d^T \nabla f(x)$$

Want d s.t. $d^T \nabla f(x) < 0$

dot product:

$$a = [a_1, \dots, a_d]$$

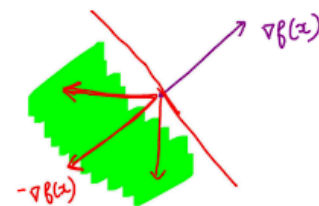
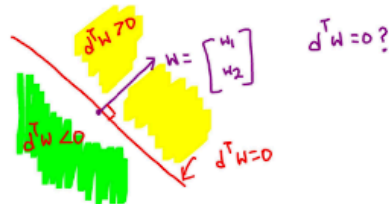
$$b = [b_1, \dots, b_d]$$

$$a^T b = a_1 b_1 + a_2 b_2 + \dots + a_d b_d$$

Note:- If $d = -\nabla f(x)$

$$(-\nabla f(x))^T (\nabla f(x)) = -\|\nabla f(x)\|_2^2 < 0$$

$$a^T a = a_1 a_1 + \dots + a_d a_d = \sum_{i=1}^d a_i^2 > 0$$



Note:- $-\nabla f(x)$ gives the "steepest" descent.

G.D is also called steepest descent.



$$\begin{array}{l} \min_{x \in \mathbb{R}^d} f(x) \\ \underline{g(x) \leq 0} \end{array}$$

$$\begin{array}{l} g(x) \\ x_1^2 + x_2^2 - 10 \\ \leq 0 \end{array}$$

QN : 1,2

Week 8 Tutorial 1

MACHINE LEARNING - FOUNDATIONS

TUTORIAL - WEEK 7

IIT Madras Online Degree

Outline

1. GRADIENT DESCENT
2. GRADIENT DESCENT VS NEWTON'S METHOD
3. COMPARATIVE EXAMPLES

GRADIENT DESCENT

<https://www.geogebra.org/m/uyvajpsy>

GRADIENT DESCENT VS NEWTON'S METHOD

Gradient Descent Update Rule

Update Rule

$$x_{n+1} = x_n - \eta f'(x_n)$$

Newton's Method's Update Rule

Update Rule

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

COMPARATIVE EXAMPLES

Example 1

Find the value of x that minimizes the following function:

$$f(x) = (3x - 9)^2$$

Traditional approach

$$f(x) = (3x - 9)^2$$

$$f'(x) = 2(3x - 9)3$$

Equating $f'(x)$ to zero, we get: $x = 3$

Putting $x = 3$ in $f(x)$, we get $f(x) = 0$.

Newton's method

$$\begin{aligned}f(x) &= (3x - 9)^2 \\f'(x) &= 2(3x - 9)3 \\f''(x) &= 18\end{aligned}$$

Newton method when $x_0 = 2$

Iteration 1:

$$\begin{aligned}x_1 &= x_0 - \frac{f'(x_0)}{f''(x_0)} \\x_1 &= 2 - \frac{18(2) - 54}{18} \\x_1 &= 3\end{aligned}$$

Only one iteration!

Iteration 2:

$$\begin{aligned}x_2 &= x_1 - \frac{f'(x_1)}{f''(x_1)} \\x_2 &= 3 - \frac{18(3) - 54}{18} \\x_2 &= 3\end{aligned}$$

Newton method when $x_0 = 1$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$
$$x_1 = 1 - \frac{18(1) - 54}{18}$$
$$x_1 = 3$$

Newton method when $x_0 = 0$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$
$$x_1 = 0 - \frac{18(0) - 54}{18}$$
$$x_1 = 3$$

Newton method when $x_0 = -100$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$
$$x_1 = -100 - \frac{18(-100) - 54}{18}$$
$$x_1 = 3$$

Newton method when $x_0 = 100$

$$x_1 = x_0 - \frac{f'(x_0)}{f''(x_0)}$$
$$x_1 = 100 - \frac{18(100) - 54}{18}$$
$$x_1 = 3$$

Gradient Descent method when $x_0 = 2$ and $\eta = 1$

$$x_1 = x_0 - \eta f'(x_0)$$
$$x_1 = 2 - 1(18(2) - 54)$$
$$x_1 = 20$$

$$x_2 = x_1 - \eta f'(x_1)$$
$$x_2 = 20 - 1(18(20) - 54)$$
$$x_2 = -286$$

Gradient Descent method when $x_0 = 2$ and $\eta = 0.05$

$$x_1 = x_0 - \eta f'(x_0)$$

$$x_1 = 2 - 0.05(18(2) - 54)$$

$$x_1 = 2.9$$

$$x_2 = x_1 - \eta f'(x_1)$$

$$x_2 = 2.9 - 0.05(18(2.9) - 54)$$

$$x_2 = 2.99$$

$$x_3 = x_2 - \eta f'(x_2)$$

$$x_3 = 2.99 - 0.05(18(2.99) - 54)$$

$$x_3 = 2.999$$

Example 2

$$f(x) = x^3 - x^2 - x + 5$$

Traditional: $f'(x) = 0$ gives $x = 1, -1/3$

Gradient Descent with $x_0 = 0.75$ and $\eta = 0.25$

GD I1: $x_1 = 0.953$

GD I2: $x_2 = 0.998$

Newton's Method with $x_0 = 0.75$

NM I1: $x_1 = 1.075$

NM I2: $x_2 = 1.004$

NM I3: $x_3 = 1.001$

Newton Method: caveats

- For higher dimensional functions, it requires computing inverse of Hessian matrix. Newton method does not work if the Hessian is not invertible.
- It may not converge. For some functions and some starting points, it may enter an infinite cycle.
- It may converge to a saddle point instead of a local minimum.
- It takes more time per iteration and is more computation and memory intensive.

PA QN : 2,3,4,13,14

GA QN : 2,3,11,13,14,15,16